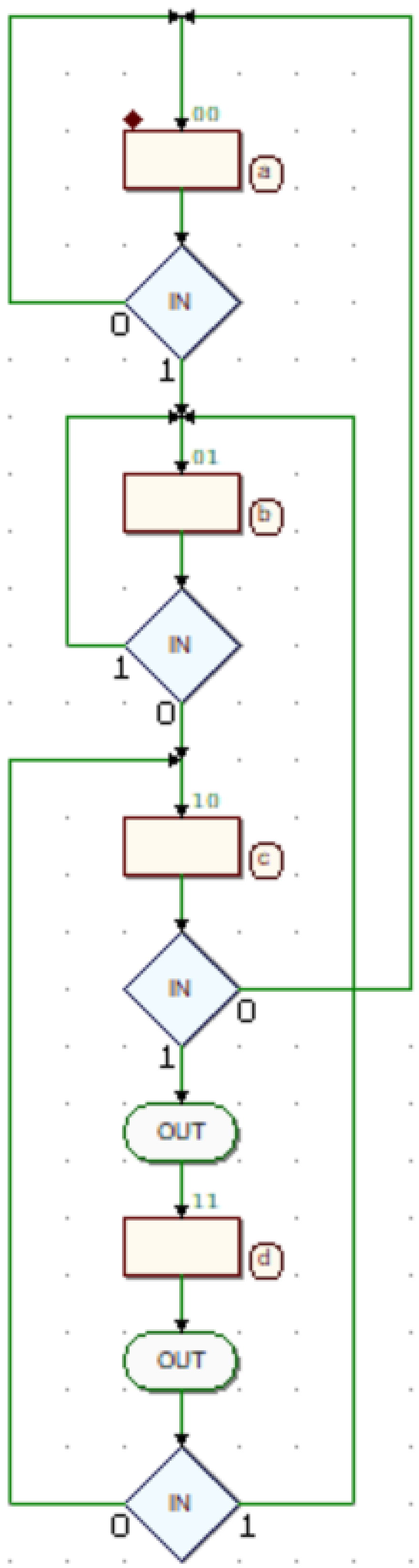
ELECTROTHON '26 DIGITAL MODULE



CHALLENGE 2: PART A

The FSM consists of 4 states labelled 'a', 'b', 'c' and 'd' in the adjacent state diagram.

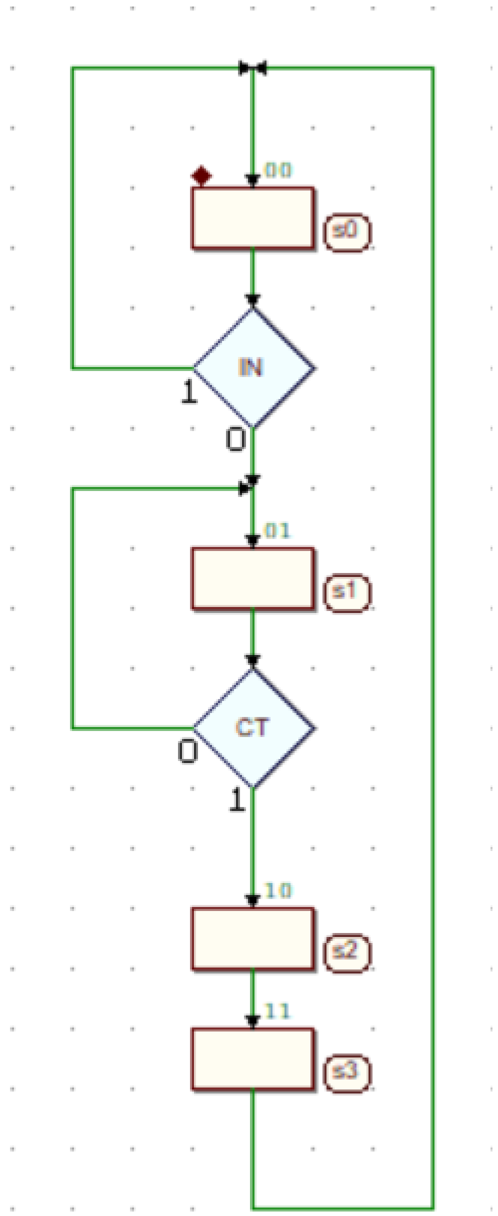
State 'a' is the initial state, and the state when there is no part of the sequence '101' present in the last four inputs.

State 'b' is the second state, where '1' is the last input but the last three inputs are not '101'.

State 'c' is the third state, where the last two inputs are '10'.

State 'd' is the fourth (and final) state, where the last three inputs are '101'.

The machine gives an output when entering/exiting state 'd'.



CHALLENGE 2: PART B

The FSM consists of 4 states labelled ‘s0’, ‘s1’, ‘s2’ and ‘s3’ in the adjacent state diagram. In the attached .v file, they correspond to S_IDLE, S_DATA, S_PARITY, and S_STOP respectively.

State ‘s0’ or S_IDLE is the state of the machine before it receives the start bit, i.e. ‘0’. Once it receives a ‘0’, it goes to state ‘s1’ or S_DATA.

S_DATA receives 8 bits of data, storing them until the count of bits stored = 8 (in the state diagram denoted as CT but different in the actual .v file). Once CT =8, it moves to ‘s2’ or S_PARITY.

S_PARITY receives one more bit as the parity bit before shifting to ‘s3’ or S_STOP.

S_STOP is the last state which lasts for only one cycle and displays all the relevant outputs such as parity_err, done and frame_err. After this cycle, it returns to S_IDLE.